MOHAMMED KAZMI

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EDUCATION

University of California, Irvine Sep 2023 - Jun 2026

Bachelor of Science, Aerospace Engineering | *Minor: Materials Science* (GPA: 3.2, cumulative incl. transfer coursework)

Coursework: Heat and Mass Transfer, Thermodynamics, Propulsion, Astronautics, Fluid Mechanics, Aerodynamics, Mechanics of Materials, Mechanical Structures, Control Systems, Flight Mechanics, Aircraft Performance, Aerospace Lab

PROFESSIONAL EXPERIENCE

HFN Builder Inc. | Mechanical Engineering Intern Jun 2024 - Sep 2024

- Produced structural and HVAC layouts in AutoCAD, improving drawing accuracy and reducing design revision cycles.
- Performed load analyses and fluid-flow simulations for piping and ventilation systems to verify code compliance and improve system reliability.
- Applied aerospace material knowledge to recommend lightweight, durable construction components, leading to cost savings and improved structural integrity.
- Evaluated insulation performance using thermodynamics and heat transfer principles, optimizing energy efficiency and reducing operational costs.
- Presented design updates in weekly reviews, incorporating engineer feedback to improve design accuracy and accelerate project approvals.

Cosumnes River College | Calculus Teaching Assistant Sep 2022 - Mar 2023

- Assisted students daily in the STEM Center across three levels of calculus, using MATLAB and C/C++ to strengthen understanding and improve academic performance.

PROJECTS

Liquid Rocket Engine Preliminary Design Tool | Python

(github.com/momokazmi1017/LiquidRocketEngineprelimdesigntool) 2026

- Developed a Python tool producing complete preliminary thrust-chamber designs; used it to design a 5 kN (1,124 lbf) LOX/ethanol regeneratively cooled engine at 25 bar (363 psi).
- Modeled chemical-equilibrium combustion (Cantera, NASA data); validated ideal specific impulse (Isp) within 0.6% of Sutton & Biblarz and c* within 1.5%.
- Ran a regenerative cooling channel trade study (Bartz heat transfer) cutting peak hot-wall temperature from 1,124 K to 758 K, below the CuCrZr limit.
- Wrote an axisymmetric method-of-characteristics (MOC) nozzle solver validated within 0.03 deg of exact theory; developed with Claude Code and verified against published data.

6-DOF Flight Simulator for a Liquid Sounding Rocket | Python

(github.com/momokazmi1017/FlightSim6DOF) 2026

- Developed a six-degree-of-freedom (6-DOF) simulator with quaternion attitude, time-varying mass/CG/inertia, atmosphere, wind, launch rail and recovery; sized a 30,000 ft rocket (max Mach 1.70).
- Validated against analytic solutions (rigid-body energy and momentum to 1e-8, rocket equation within 0.2%) and ran a 500-run Monte Carlo dispersion: apogee 30,080 +/- 2,330 ft, all runs met requirements.
- Fin flutter analysis (NACA TN 4197) caught undersized fins (margin 1.32); redesign to 1/4 in G10 raised margin to 2.01.

Boeing 787-9 Class Twin-Engine Transport | Senior Design Project | UC Irvine Jan 2026 - Mar 2026

- Led the conceptual and preliminary design of a 787-9 class transport, executing the full sizing loop from mission requirements through weight, aerodynamic and performance convergence.
- Built a MATLAB sizing and performance model integrating atmosphere, aerodynamics and propulsion to evaluate takeoff, climb, cruise and range; calibrated the drag polar against published data.
- Conducted trade studies across wing geometry, weight and thrust to optimize fuel efficiency and meet range and field-length targets.

Oblique Wing Wind Tunnel Analysis | Team Lead | UC Irvine Mar 2024 - Jun 2024

- Designed and fabricated an oblique wing (NACA 4415) and led wind tunnel testing at varying sweep and attack angles from 5 to 30 m/s.
- Set up and calibrated data acquisition (LabVIEW, pressure transducer, pitot tube) to collect lift, drag and torque; analyzed results in MATLAB to identify sweep angles maximizing L/D.

Noise Abatement Optimization for Boeing 747-400 | UC Irvine Jan 2025 - Mar 2025

- Modeled 747-400 takeoff at LAX in MATLAB and proposed a modified NADP1 climb profile that reduced sound exposure level by up to 22% at 5 miles from the runway.

1996 Jeep Cherokee | Personal Automotive Project Aug 2020 - Present

- Restored and upgraded drivetrain, suspension, brakes, intake/exhaust and fuel systems; diagnosed electrical, mechanical and structural faults using OBD-II tools and a multimeter.

SKILLS

Analysis and Simulation: Heat Transfer and Thermodynamics, Propulsion, Compressible Flow, Flight Performance and Trajectory Modeling, Orbital Mechanics and Delta-V, FEA (Ansys), Autodesk CFD, Structural/Load Analysis, Control Systems, Monte Carlo Analysis

Software and Tools: MATLAB, Simulink, Python, C/C++, SolidWorks, CATIA, Creo, AutoCAD, Ansys STK, LabVIEW, Git, AI-Assisted Development (Claude Code)

Design and Test: Trade Studies, Model Validation, GD&T, Tolerance Analysis, DFMEA, Data Acquisition, Thermocouples, Accelerometers, Vibration Testing, Technical Writing